

HROC • NCAN • DEEP-LEARNING TRACK

Four animals in. What's real, and what isn't.

Behaviour is solid. The cortical drift did not survive its own control.

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SINCE LAST MEETING

What changed

1

4 animals processed

9, 10, 11 down-conditioned; 12 up. Directions read straight from the experimenters' logs.

2

Per-day analysis

Timestamps now in the data, so drift is measured per day / 5-day block, not per phase.

3

The null control

Your sham test: split baseline-only data as a fake experiment and see if it 'drifts'.

~1.6 million trials decoded across the four animals.

Three terms for the next few slides



H / M ratio

The reflex size divided by the direct muscle response. We never use raw reflex size, because it depends on how hard you stimulate.



Drift

How far the average brain state has moved from where it started at baseline. Bigger number means the representation changed more.

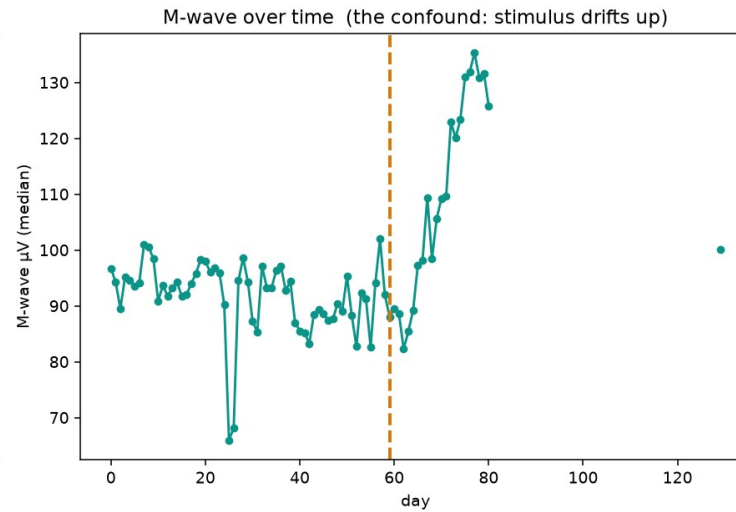
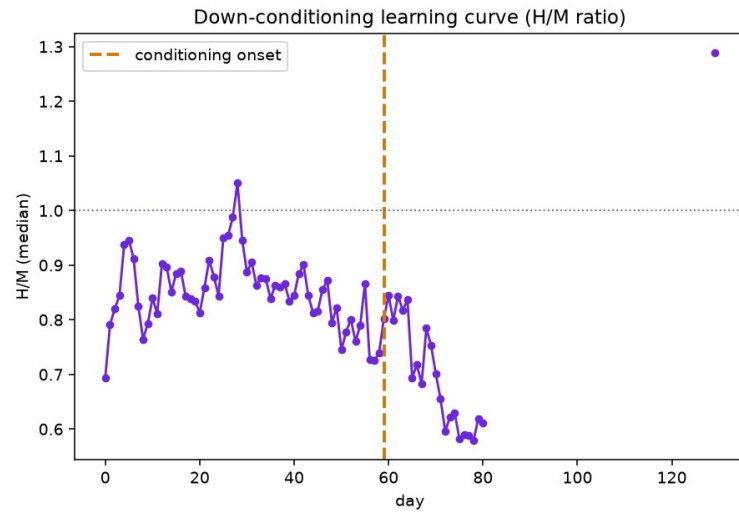


The null control

Take baseline-only data, pretend half of it was 'conditioning', and run the same analysis. There was no real conditioning, so it should show ~zero drift.

RESULT 1 • THE BEHAVIOUR

Down-conditioning worked



Animal 11

H/M sits flat around 0.87 through baseline, then falls steadily to about 0.60 once conditioning starts. Animal 9 shows the same pattern.

This is the textbook curve, and it's the most solid thing we have.

The null control, and why it matters

The idea

Take ONLY the baseline recordings, before any conditioning happened. Pretend the first part is 'baseline' and the rest is 'conditioning'. Run the identical drift analysis.

Nothing happened in that data, so an honest method must report ~zero drift.

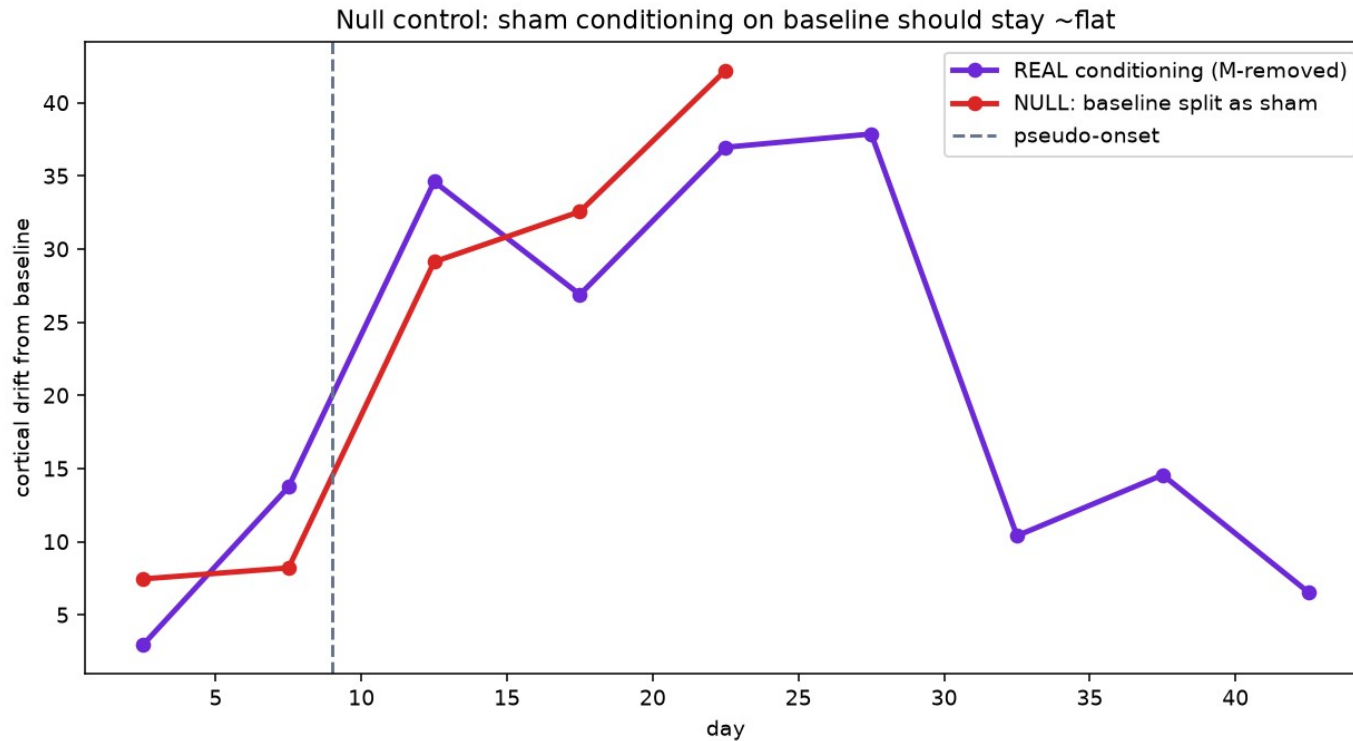
What we found

The sham 'drifts' just as much as real conditioning in 3 of the 4 animals.

So most of what we were calling a cortical effect is slow drift in the recording itself over weeks, not learning.

THE NULL CONTROL, IN ONE PICTURE

Animal 9: the sham drifts as much as the real thing



Red = fake experiment

Baseline-only data, no conditioning at all. It climbs as high as the purple real-conditioning line.

Also note the purple line rises during baseline and falls during the strongest learning. It runs opposite to the behaviour.

ALL FOUR ANIMALS

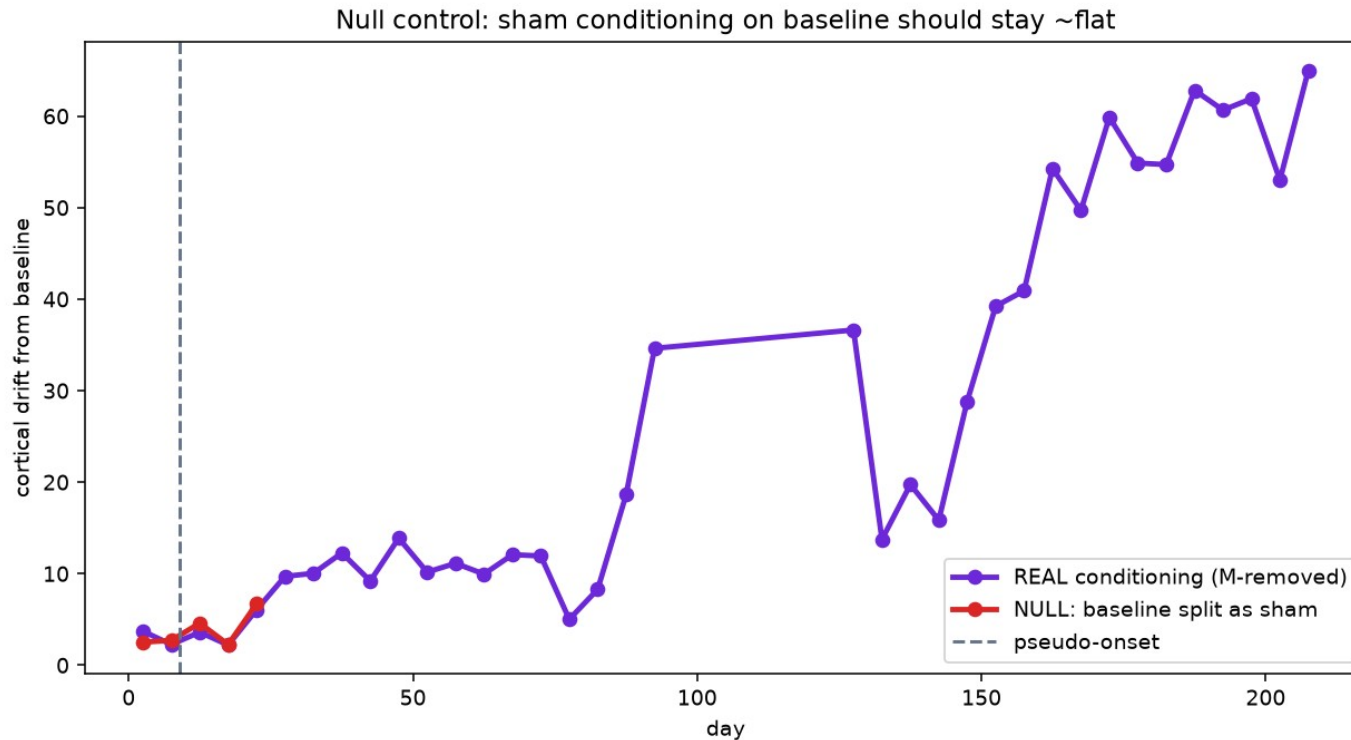
The scorecard

Animal	Direction	Behaviour	Null control
A9	down	Worked (H/M falls)	Fails
A10	down	Complicated (40-day gap)	PASSES
A11	down	Worked (H/M falls)	Fails
A12	up	Weak responder	Fails

Down-conditioning is reproducible. The cortical drift is not — only one animal survives its own control.

THE ONE EXCEPTION

Animal 10 passed the null control



Real 30.6 vs sham 6.7

Here the real conditioning drift is about 4.6x the sham. This is the one animal where the cortical change is not explained by recording drift.

Worth chasing. But one animal out of four is a lead, not a result.

Why one animal can't settle this

In a single animal, conditioning and recording drift are both slow changes over weeks. They are mathematically tangled, so no single-animal analysis can separate them.

What is solid

- Decoding + pipeline across 4 animals
- Down-conditioning quantified with H/M
- The control methodology itself

What is not

- A conditioning-specific cortical drift
- Any up-vs-down cortical contrast
- Pooling animals (results disagree)

What would make a cortical claim believable

A cortical signature has to survive all five of these. Right now we clearly pass one.

1

Null control

sham on baseline shows ~zero

only 1 of 4

2

M-wave control

not driven by stimulus size

cleared

3

Pre-stimulus control

not just arousal / background

not built yet

4

Direction contrast

up and down differ or flip

need a strong up animal

5

Tracks behaviour

follows the H/M learning curve

not yet

PLAN

Next few weeks

1

Per-animal clean-up

Handle recording gaps, set M/H windows from each animal's own average.

2

Pooled encoder

Retrain the representation on all animals, not just animal 9.

3

Pre-stimulus control

Add background excitability alongside the M-wave.

4

Chase animal 10

Is its surviving drift real, or luck?

5

A strong up animal

The single highest-value thing to add.

We're deliberately not rushing a draft. Better to clear the bar than publish a confounded effect.

WHERE I NEED YOUR READ

Four questions

1 Is the null-control read right?

Baseline sham drifts as much as conditioning. Does that convince you the drift is recording, not learning?

2 Animal 10

It passed. Worth chasing, or likely a fluke given 3 of 4 failed?

3 A strong up animal

Which animal in the drive is most likely a solid up-conditioning responder?

4 The bar

Are the five controls the right bar, or would you add or drop one?

Everything (code, figures, per-animal results) is in the repo.